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How does AI drive branding? Towards an integrated theoretical framework for AI-driven branding

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ABSTRACT

Brands are transforming with the acceptance, adoption, and use of artificial intelligence (AI) technologies. AI builds dynamic consumer-brand relationships that help build sustainable businesses for brands to thrive. The brands sustain in competitive markets by crafting engaging brand narratives that holistically influence consumer purchase decisions. The present study comprehensively reviews the theoretical underpinnings of AI-driven branding from the extant literature. The thematic analysis of theoretical premises instrumentalises developing a novel integrated theoretical framework. The intricate analysis of the constellation of theories and the simultaneous amalgamation of concepts allows for a unified viewpoint. The integrated theoretical framework offers an in-depth overview of the drivers, mediators, moderators, and consequences of AI-driven branding. The integrated theoretical framework extends the current theoretical understanding of consumer-brand relational dynamics with AI that practitioners may use to sustainably transform their brands.

1. Introduction

Artificial Intelligence (AI) is the recent technological disruptor responsible for transforming marketing strategies in mammoth proportions (Verma et al., 2021). The academic contributions towards deciphering the sustainable impact of AI on marketing are significant and continue to garner interest. The nexus of Branding and AI is witnessing a colossal rise in the scholarly marketing research domain. The marketplace is cognizant of the rampant technophilia that brings both brands and consumers a sense of spirited gratification (Belk, 2021) and cultivates sustainable business relationships (Goralski & Tan, 2020). AI goes beyond just a progression of statistics and data distributions. Digitisation is prevalent across all fields in the corporate arena, including human resource management (Votto et al., 2021), tourism management (Verma et al., 2022), civic governance (Verma, 2022), sustainability (Nishant et al., 2020) etc. Predictive capabilities of AI fuel effective, sustainable and consistent brand strategy. Personalised campaigns, collaborative initiatives, and predictive modelling offer incomparable insights for marketing researchers and practitioners (Campbell & Price, 2021). AI-driven branding facilitates the connection between innovation and sustainability due to the mutually symbiotic relationship between man and machine (Vaio et al., 2020).

The AI-driven branding literature is growing exponentially considering the diverse applications of AI in brand activities (Davenport et al., 2020; Huang & Rust, 2021; Martin & Murphy, 2017; Mariani & Baggio, 2022; Hermann, 2022; Kaplan & Haenlein, 2020; Dew et al., 2022; Verma et al., 2021; Mariani et al., 2023; Lin & Wu, 2023). The AI-driven branding literature houses a myriad of theories rooted in different disciplines to explain the research topic. For example, the norm as a construct has been used widely to explain AI-dtheriven branding from different theoretical perspectives. Dwivedi et al. (2021) used the theory of planned behaviour to explain the role of subjective norms in social influence for a positive response to AI-driven branding. Cantallops and Salvi (2014) used subjective norms from the Technology Acceptance Model perspective to explain the user acceptance of AI in brand interactions. Kim et al. (2023) used social capital theory to the reciprocal norm for liking social media brand activities. Chan et al. (2022) used presumption theory to explain the role of norm-based message tunning for garnering brand support in social media initiatives. As evident, multiple theoretical lenses are applied across a plethora of research references to explain the role of norms in AI-driven branding, leading to theoretical and interdisciplinary confusion. The problem lies in the conflict of choice among various theories across disciplines. There is an urgent need to schematically and thematically consolidate the theories used in the extant literature to offer a unified view of AI-driven branding. Without such an analysis, the successful

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progression of the AI-driven branding domain would be a protracted task (Mustak et al., 2021). Moreover, the void of a unified theoretical perspective impedes future research potential and limits the advancement of the domain. Hence, a theoretical review would act as a catalyst to stimulate the future research agenda and propagate the AI-Brand dominion. A holistic and unified theoretical review through schematic, thematic, and content analysis will help to build an integrated perspective to guide future researchers and practitioners. Thus, the present study fills the research gap by offering an integrated theoretical framework to explain AI-driven branding.

Previous researchers have presented literature reviews from varied perspectives to capture a synthesised view of the literary contributions of AI-driven branding. A review by Varsha et al. (2021) conducted a bibliometric study on 117 research articles from 1982 to 2019 to assess the performance of science actors and map the intellectual structure of the topic of AI-driven branding. Dropulić et al. (2022) conducted a systematic literature review with 82 research articles from 2011 to 2021, applying a narrative style of review presentation. Vaio et al. (2020) conducted a systematic literature review assessing how AI impacts the creation of sustainable business models for brands, with a dataset of 73 papers from 1990 to 2019. A literature review of 1451 articles applying a mixed-method approach (combining topic-modeling and content analysis) is done on AI maturity and consumer readiness using a socio-technical frame of reference (Akbarighatar et al., 2023).

All of the previous studies have focussed on describing the past literature and examining the dimensionality of the domain of AI in branding. However, a theoretical lens with a unified viewpoint is missing. The previous reviews have consolidated the academic contributions and thematically described the same. However, a comprehensive account of the theoretical underpinnings used in AI-driven branding and the subsequent integration of the theoretical lenses is an evident gap in the literature. Hence, the present theoretical review resolves the unaddressed problem of a unified theoretical viewpoint that promises future researchers the ease of reviewing the theories used in AI-driven branding literature all at once.

Additionally, previous researchers have selected papers till 2021 and included less than 250 documents. The present study has considered 414 research articles published in Scopus-indexed journals between 2000 and 2022. Thereby, assuring a detailed and systematic analysis in recency, quality of articles, and rigour. Thus, the present review attempts to widen the horizons of brand research by systematically synthesising, reviewing, and unifying the theories in the past literature. The present study unfolds multiple facets of AI in branding for advancing the research domain by identifying a constellation of prominent theories to build a unified theoretical framework that highlights the AI's precursors, mediators, moderators, and outcomes in the branding phenomenon. The review is an integrated, conceptually rich thematic analysis of the research domain's theoretical structure (underlying theories, emerging themes, and conceptual definitions). Given the above discussion, the present study aims to resolve the definitional ambiguities and shape the topic by developing an integrated theoretical view.

The research questions of this study are as follows:

RQ 1. What is the theoretical foundation of AI-driven branding?

RQ 2. What is the integrated theoretical framework for explaining AI-Driven branding? What are the drivers, consequences, mediators, and moderators of this phenomenon?

RQ 3. What are the implications of an integrated theoretical framework for theory, practice, and policy?

The present integrative review with a theoretical lens aims to develop an integrated theoretical framework to propose propositions to guide theory and practice. Along the discovery-justification continuum resonating with knowledge development (Hanson, 1958), the present study validates what is known and accumulates the theories used to explain the phenomenon of AI in branding. Thus, the study contributes to the theoretical development by corroborating and summarising what is known with an integrated viewpoint (MacInnis, 2011). Based on a systematic and structured review encompassing thematic and content analysis, the present study proposes an integrated theoretical framework to address the research questions. The extant theoretical review with underlying theoretical underpinnings is analysed metaphorically like an architectural plan wherein the researcher plays the role of an architect to create an integrative framework (MacInnis, 2011).

The paper is structured comprising seven sections. Section two presents the theoretical background, and section three details the research methodology. Section four outlines the prominent theories used in extant literature to explain the role of AI in branding and answer RQ 1. Section five integrates the theoretical premises using thematic and content analysis to develop an integrated framework and addresses RQ 2. Section six unfolds implications for theory, practice, and policy to address RQ3. Section seven concludes the contribution of the paper.

2. Theoretical background

2.1. Branding and AI

AI is defined in varied ways to ultimately insinuate the splendour of technology, characterised by personalisation and convenience. AI facilitates three broad functions: "interpret external data correctly, learn from such data, and exhibit flexible adaptation" (Kaplan & Haenlein, 2019, p. 17). AI in branding has evolved under the purview of three dimensions – *Level of Intelligence, Task Type (task automation or context awareness), and AI in Robotics* (Davenport et al., 2020). AI is "manifested by machines that exhibit aspects of human intelligence" (Huang & Rust, 2018, p. 155). The extant literature presents either all-encompassing definitions (Shieber, 2004; Brooks, 1991; Russell & Norvig, 2010; Legg & Hutter, 2007) or too-restrictive definitions (Goertzel, 2015; Thórisson et al., 2015; Kurzweil, 2005). According to De Bruyn et al. (2020, p.3), an appropriate and definite definition of AI is "machines that mimic human intelligence in tasks such as learning, planning, and problem-solving through the higher level, autonomous knowledge creation."

A "Brand" is defined in a myriad of ways owing to the multiple nuanced perspectives presented in the extant literature. The American Marketing Association (AMA) described a brand as a logo in 1960. The AMA redefined " Brand " in 2007 as "A name, term, design, symbol, or any other feature that identifies the seller's good or services as distinct from those of other sellers." (AMA, 2007) A brand is considered a legal statement of ownership (Crainer, 1985), offering corporate identity (Van Riel & Balmer, 1997; Varadarajan et al., 2006), acting as a memory shortcut to building mental connections (Keller, 2003; Jacoby et al., 1977) and for some brands, creating an emotional association with consumers (Kapferer, 2008). Brands present an identity framework comprising six unique aspects (Kapferer, 1994), namely, culture (O'Reilly, 2005), personality (Aaker, 1997; Zinkhan et al., 1996; Azoulay & Kapferer, 2003; Goodyear, 1993), self-projection, physique, reflection, and relationship (Woodward, 1991; Blackston, 1992). Brands and their associated branding activities are an evolving phenomenon that paves the way for successful differentiation in a competitive and fickle marketplace.

By displaying facets of human intelligence, AI is dynamically transforming the branding landscape through inventive breakthroughs, creative content, immersive experiences (Kar &Varsha, 2023) and diverse communication channels (Haenlien & Kaplan, 2019). According to Marinchak et al. (2018), AI is applied to effectively segment and target customers, create appropriate brand advertising content and predict consumer preferences to offer a personalised experience. Extant literature depicts how advanced AI-enabled applications are being embedded into branding strategies, brand engagement plans and brand interactions by way of virtual assistants (Jones, 2022; Maroufkhani et al., 2022; Loureiro & Bilro, 2022), chatbots (Lin & Wu, 2023; Chung et al.,

2020; Arjun et al., 2021), robotic applications (Chan & Tung, 2019; Hwang et al., 2021; Davenport et al., 2020), augmented reality platforms (Tan et al., 2022; Rauschnabel et al., 2019; Scholz & Duffy, 2018), virtual reality interfaces (Bigne et al., 2016; Rather et al., 2023; Regt et al., 2021), branded avatar interactions (Belk, 2014; Miao et al., 2022; Lin et al., 2021) and immersive experiences (Kozinets, 2023; Ardura & Artola, 2019; Wongkitrungrueng & Suprawan, 2023; Kar & Varsha, 2023). Furthermore, as AI reimagines brand management, several facets of branding are being influenced and augmented by the forces of AI technologies.

2.2. Brand equity and AI

In the age of digitisation, brand equity is reflected by the "added value" (Farquhar, 1989, p. 24) rendered by AI-driven technologies. According to Keller (1993, p. 1-22), brand equity is the "differential effect that brand knowledge" creates on consumer perceptions towards the branded efforts. Predictive capabilities, social media analytics and natural language processing (NLP) help brands deliver differentiated offerings, thus, achieving high brand equity to stay competitively strong (Rajagopal, 2020). AI impacts brand equity by reflecting in actionable outcomes like brand volume, profit margins, awareness and advocacy (Oh et al., 2020). AI tools like data analytics, sentiment analytics, machine learning and deep learning are influencing the three dimensions of brand equity, namely customer mindsets (e.g., attitude, loyalty, attachment), product-market outcomes (e.g., price premium, market share) and brand deemed a financial asset (Ailawadi & Keller, 2004; Oh et al., 2020). User-generated content (UGC) on social media platforms or brand online communities contributes to building or attenuating the brand's image, thus impacting the brand's overall equity (Kuksov et al., 2013). Ergo, brands use AI techniques to obtain, analyse and predict consumer opinions using the learnings and patterns emerging from the big data (Dwivedi et al., 2023).

2.3. Brand sustainability and AI

Implementing and integrating sustainable branding and operations functions enhances sustainable business practices (Kumar & Christodoulopoulou, 2014). Brands can be used as strategic means to achieve sustainable processes with the help of AI acting as the technological facilitator (Backhaus et al., 2011). The study by Nishant et al. (2020), highlights the adverse impacts of AI implementation on building sustainable brand businesses including growing cybersecurity risks, consumer vulnerability to data breaches, and difficulty measuring the impact of technological strategies. Rapid developments in AI technologies are changing the patterns of consumption and production owing to Knowledge Management Systems (KMS) emerging as an effective pathway for brands to create sustainable business models (Vaio et al., 2020). AI assists in augmenting brand performance by implementing eco-innovation at the product, process and organisational levels (Larbi-Siaw et al., 2022). AI technologies tread on futuristic ideals to create a disrupting or enhancing effect on brand initiatives such that the competitive advantage is sustained (Goralski & Tan, 2020).

2.4. Brand personality and AI

Brand personality revolves around identifying the type of personality a brand represents and the degree of personality congruence there is between a consumer and the brand (Zinkhan et al., 1996). In the AI era, it is important to understand and attribute humanlike personality traits to brands as there is a direct influence on consumers' actions and the relationship they share with the brand. In 1997, Aaker coined the term brand personality and defined it as the set of human traits attributed to a brand. With the onset of AI-driven applications, brand personality is being applied to an influx of technological contexts to explain consumer responses to brand initiatives. With a contextual shift of brand personality lens towards the rampant digital sphere (Torres & Augusto, 2019), extant literature is surveying how consumers are attaching anthropomorphic characteristics to brand entities during their online interactions on social media platforms (Machado et al., 2019), virtual conversational agents (Youn & Jin, 2021; Vernuccio et al., 2021) and brand owned media (websites) (Shobeiri et al., 2015). From the perspective of differentiation, building and conveying the right kind of brand personality can generate a positive UGC as opposed to focusing only on functional brand benefits (Chernatony, 2010).

2.5. Brand elements and AI

Brand elements like the logo, tagline, colours and graphics impact brand visibility as it is present on product packaging, brand merchandise, advertising content, brand websites and social media platforms. Brand elements are the "name, term, design, symbol, or a combination of them" for easier identification and effective differentiation (American Marketing Association, 1960). Researchers have challenged the definition for being too narrow in perspective as intangible brand elements (mission, vision and purpose) have not been considered (Crainer, 1995). The brand logo and other elements convey the spirit of the brand. Thus, with the emergence of AI, brands can use advanced technologies to design impactful brand elements (Dew et al., 2022) or use AI techniques to gauge consumers' perceptions of a newly designed/redesigned logo (Capatina et al., 2020). On social media, consumer photos with the brand logo or signage present depict brand loyalty, advocacy, love and positive word-of-mouth (Varsha et al., 2021). AI techniques like machine learning algorithms and artificial neural networks can help understand why consumers tap into brand profiles to get the brand's attention and display loyalty to the brand (Kapidzic, 2013; Skotis et al., 2023).

2.6. Brand relationships and AI

Consumer-brand relationships (CBR) comprise six key elements: partner quality, intimacy, behavioural interdependence, personal commitment, self-concept connections and love (Fournier, 1998). All six dimensions were introduced in the seminal study by Fournier (1998), wherein the Brand Relationship Quality Model (BRQ Model) was conceptualised. CBR has a significant impact on a brand's successful existence in the marketplace as it enhances the brand's repute, builds a loyal customer base and helps the brand survive in a largely competitive ecosystem (Sweeney & Chew, 2002). In the age of advanced technologies led by AI, brand practitioners aim to leverage the conversational abilities of AI agents like chatbots, digital avatars, virtual assistants and physical-embodied robots to forge long-lasting CBR (Lin & Wu, 2022). The personalised conversations and prompt customer service rendered by AI-driven conversational agents build positive consumer perceptions and need gratification. Thus, effective social interactions lead to the strengthening of brand relationships and consequently increased purchase intention from loyal consumers (Lin & Wu, 2023). Cheng and Jiang (2022) highlight how chatbot marketing initiatives indirectly impact CBR as consumers consider the interaction quality, information availability, accessibility and personalisation capabilities (Nguyen et al., 2021). The quality of AI services impacts the relationship between brands and consumers. With high-quality AI, consumers are pleased with the reliability, speed and customisation, thus directly positively impacting their perception of the brand and building a strong bond. The quality of consumer-brand relationships impacts the sustainability of brand efforts in engagement and visibility. Factors like trust moderate, whereas psychological barriers and product usage doubts mediate the relationship between anthropomorphic AI assistants and CBR (Jham et al., 2023). Thus, automated AI agents support brands in creating or enhancing CBR through programmed predictive capabilities that personalise and delight consumers.

3. Research methodology

According to Tranfield et al. (2003), a literature review performs a two-fold task. A review scientifically maps the existing research domain and evaluates a future research direction that can further strengthen the knowledge on the topic. A review is characterised by identifying and synthesising relevant literature to streamline, examine, and answer specific research questions (Palmatier et al., 2018). The present study adopts both current and rigorous methods, the bibliometric analysis technique (Donthu et al., 2021) and the Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR) Protocol (Paul et al., 2021). Comprehensive analysis with the systematic integration of 414 documents drives the thematic analysis to develop a holistic theoretical viewpoint. The present theoretical review retrieves extant theories cited in relevant literature to establish a unified framework for enabling a cohesive view (Venkatesh et al., 2003). Such reviews act as the steering wheel, directing and progressing the research domain on the right path to reach the desired potential of understanding the phenomenon

The present study maps the intellectual structure of AI-enabled branding based on the Scientific Procedures and Rationales for Systematic Literature Reviews (SPAR-4-SLR) Protocol (Paul et al., 2021). The SPAR-4-SLR protocol allows us to present an integrated overview of the current state of knowledge in three stages: assembling, arranging, and assessing. The 414 documents were retrieved from the comprehensive citation database, Scopus by creating a query string that holds keywords like "artificial intelligence", "brand", "branding", "machine learning", "deep learning", "text analytics", "data mining" and "predictive analytics". In the first phase of the search mission which is Assembling, we obtained 4000 documents. In the second phase, Arranging, certain inclusion and exclusion criterias were considered and applied. To adequately classify the data we applied filters pertaining to Language (English), Time Frame (2000 – 2022), Source Type (Indexed Reputed Journals) and Document Type (Published Peer-Reviewed Research Papers). A combination of the system and manual filtering led to the retrieval of 414 documents for further analysis.

A thorough review of 414 documents from 2000 to September 2022 forms the basis of the bibliographic cluster analysis. Louvain algorithms-based bibliographic network analysis (Zupic & Cater, 2015) on VosViewer identifies the thematic clusters, building the research domain's conceptual foundation. Extant literature suggested various community detection algorithms with different types and scales of complex networks (Zhang et al., 2018b). Christensen et al. (2020) compared eight different community detection algorithms – Walktrap, Infomap, Fast-greedy, Louvain, Leading Eigenvalue, Label Propagation, Spinglass, and Edge Betweenness for performance evaluation and found the Louvain algorithm superior to others when nodes proximity was measured on Euclidean distance. Other researchers like Brusco et al. (2022) and Gates et al. (2016) also attempted similar studies to compare the community detection algorithm and found the advantages of using the Louvain algorithm. Louvain Algorithm (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008) works in two phases. The first phase works on the modularity optimization process (local moving of nodes), followed by the network aggregation process. Louvain algorithms discern

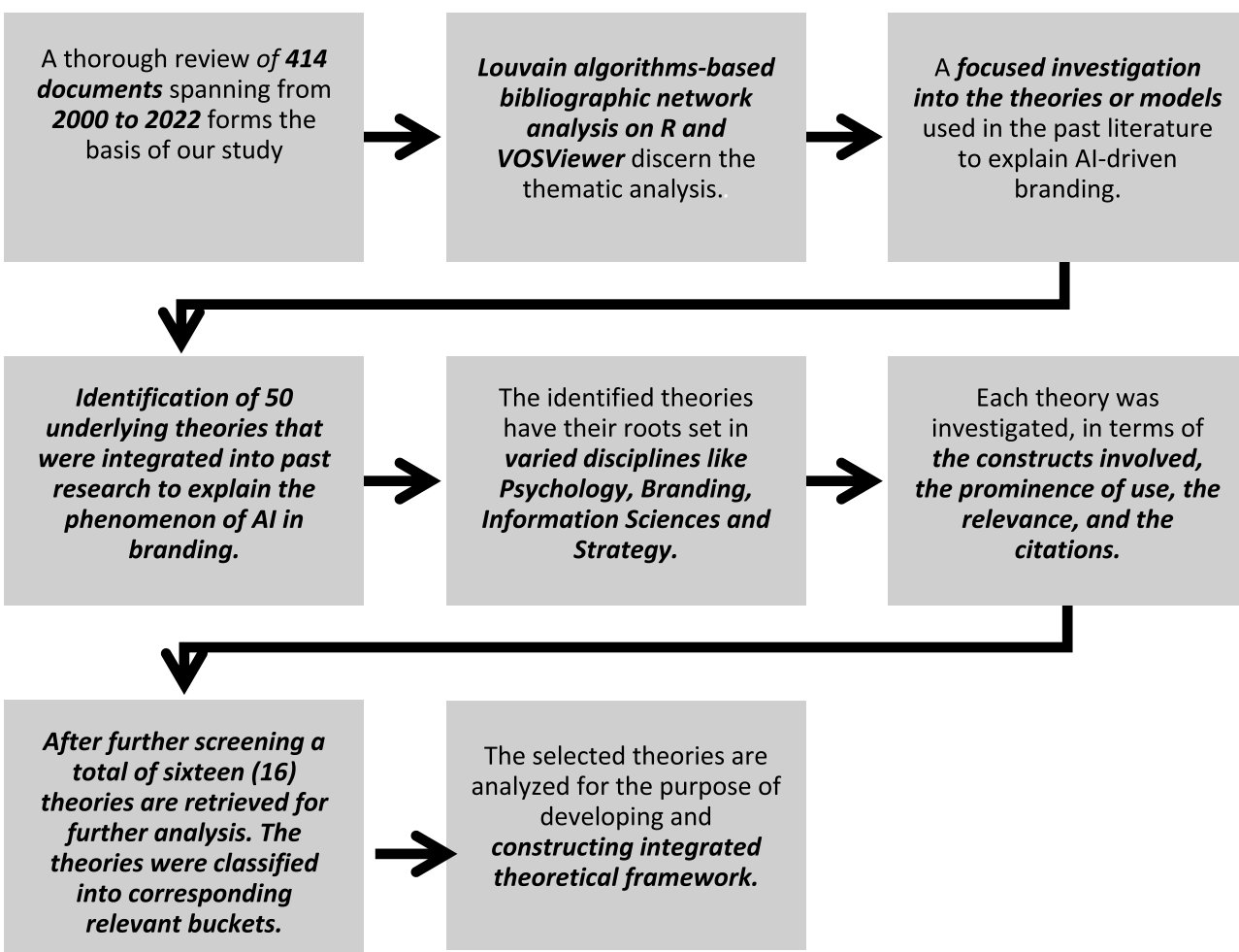

Fig. 1. Methodology flowchart.

well-defined communities based on hierarchical structuring of communities at a faster rate and hence, is a preferred choice for network analysis.

Ergo, the thematic analysis of the eight bibliometric clusters discerns the dimensionality of the research domain and paves the way for future research agendas. The flowchart representation of the methodology adopted for building the integrated theoretical framework is presented in Fig. 1.

The focused thematic analysis of the theories or models used in the past literature to explain branding resulted in identifying 50 underlying theories integrated into scholarly articles to explicate and represent the topic of interest. The fifty (50) underlying theories are rooted in various disciplines ranging from Psychology, Branding (Marketing), and Information Sciences to Strategy. Based on the thematic and content analysis, these theories were classified into corresponding relevant buckets. Each theory is thematically examined regarding the constructs involved, the prominence of use, the relevance, and the citations.

After further screening and scrutiny, sixteen (16) theories are retrieved for further analysis. The selected theories analysed to develop and construct the integrated theoretical framework are mentioned below in Table 1. The premise, the associated concepts, and the critical aspects of each theory are captured in the upcoming sections. These sixteen (16) theories are used to build the integrated theoretical framework that incorporates all significant elements under one framework. All the constructs act as building blocks stacked together to create a strong theoretical foundation. The Psychology theories lend their behavioural explanation prowess. The information science theories assist in explaining the factors that influence the acceptance and use of modern technologies. The branding (marketing) theories bring forth the dimensions of branding that are being transformed or influenced by the AI algorithmic machinery and act as the missing piece of the puzzle.

4. Theoretical underpinnings

The present section highlights and describes the intricacies of the proposed integrated theoretical framework. There is a rich diversity in the theories originating from multiple disciplines to explain, comprehend and establish the nuances of AI-driven branding. Sixteen prominent theories were considered, seven from Psychology, six from Branding, and three from Information Sciences. This integrated theoretical framework is formulated on the solid foundation of the conceptual and operational similarities across all sixteen theories. The constructs present in these note-worthy theories have been thematically reviewed, understood, and presented as an antecedent, moderators, mediators, or outcomes depending on their role in this story of how AI influences branding. Fig. 2. represents the unified view of all the drivers, mediators, moderators, and outcomes of AI-driven branding. The present study describes the relational dynamics by presenting an integrated theoretical framework. The holistic view comprising all possible relationships among the aligned constructs explains the relational dynamics in unison. The authors explain each section of the model in detail below.

The section additionally acquaints the readers with theoretically rich propositions that will direct future research down the right path. The propositions are tentative statements that explore the possible relationships between the constructs that emerge from the listed prominent theories. The constructs emerge from varied theories that are deeply rooted in established disciplines. Time-honoured disciplines like Psychology, Marketing, and Information Sciences have contributed vastly to the growth and development of knowledge across domains. The propositions elucidated in the present study enable researchers to explore the relationship of constructs with an interdisciplinary scope. The opportunity to delve into the inter-relationships of constructs used in theories of different disciplines gives rise to fresh and novel conversations. The Research Propositions (RP) have been mentioned in the following discussion. The list of the prominent theories that have enabled the creation of the integrated theoretical framework has been mentioned below in Table 2. A brief yet impactful glance at the theoretical lenses and how each theory has helped explain AI in branding has been discussed in the following section. The evolution of the theories across time (2002-2021) has been mapped in Table 1.

To fuel a better understanding of the theoretical background, the present study showcases and describes the sixteen dominant theories from the literature that help illustrate and delineate AI in branding. Each of the theories helps in understanding how AI supports the possibility of building, driving and sustaining brand success.

4.1. Social network theory

The Social Network Theory highlights the processes of communicating information and social relationships' role in transmission. The theory explores how personal, or media influence should be steered for facilitating attitudinal or behavioural change. The theory has been revolutionary in understanding and explaining media effects research and advertising impact on brand and social media behaviours from its origination in 1960 till date. The growing use of network analytical techniques in various contexts, including AI-driven branding, has enabled brand managers to gauge the nuances of various networks the brand shares. The three key network concepts that enable the theory to explain the phenomenon are centrality, cohesion, and structural equivalence. Structural centrality is represented by three unique measures, degree, closeness, and betweenness (Freeman et al., 1979). Network cohesion measures the degree of interconnections for discovering subgroups within larger social networks (Burt, 1987). Theoretical insights like the theory of weak ties (Granovetter, 1973) and the theory of diffusion of innovations (Burt, 1987) contribute to strengthening the understanding of how information flows, its mediation, and the outcomes.

The Social Network Theory provides reasoning for assessing the interrelational characteristics needed to build a fruitful social media plan. The theory posits the relevance of the structural features of the consumer's relationship with the brand or the platform. The number of

Table 1
The evolution of theories in AI-in-branding literature.

	SNT	SIT	TPB	SLT	SCT	TCD	PRT	CCT	MRT	TCT	CBRT	BPT	UGT	TAM	UTAUT	DIT
2021			1		1					1				1	1	
2020	1						1							1	2	
2019			1												1	
2018						1		1			1		3			
2016												1				1
2015				1							1					
2014		1	1			1			1		1			1		
2013										1						
2011	1												1			
2010													1			
2006																1
2002		1	1								1					

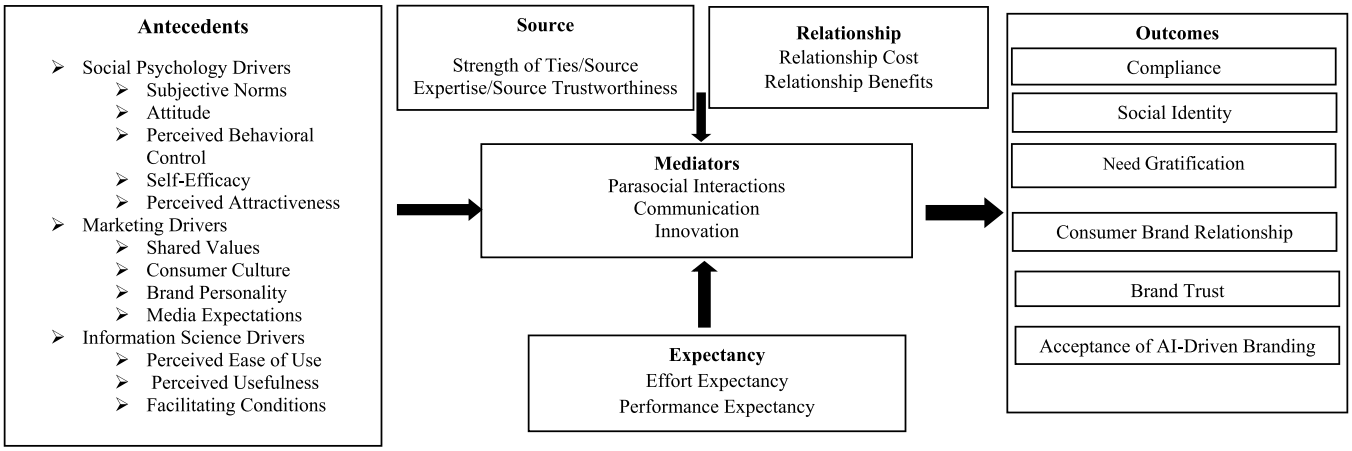

Fig. 2. Integrated theoretical framework.

Table 2
Prominent theories used for explaining the role of AI in branding.

Field	Theory	Origin	Year	Citations	Highly Cited AI in Branding Paper	Year	Citations
Psychology	**Social Network Theory (SNT)**	Granovetter, 1973	1973	66317	Kietzmann et al. 2011	2011	7333
Psychology	**Social Identity Theory (SIT)**	Tajfel et al., 1979	1979	30066	Bagozzi & Dholakia, 2002	2002	2578
Psychology	**Theory of Planned Behaviour (TPB)**	Ajzen, 1991	1991	108497	Cantallops and Salvi, 2014	2014	1029
Psychology	**Social Cognitive Theory (SCT)**	Bandura, 1986	1977	111408	Ngai et al., 2015	2015	965
Psychology	**Source Credibility Theory (SCT)**	Hovland et al., 1953	1953	10241	Dwivedi et al., 2021	2021	561
Social Psychology	**Theory of Cognitive Dissonance (TCD)**	Festinger, 1957	1957	56407	Colicev et al., 2018	2018	235
Psychology	**Para-Social Relationships Theory (PSRT)**	Horton and Richard Wohl, 1956	1956	5312	Pitardi and Marriott, 2021	2020	74
Marketing	**Consumer culture Theory (CCT)**	Arnould and Thompson, 2005	2005	5339	Brodie et al., 2013; Hollebeek et al., 2014	2014	2535
Marketing	**Media Richness Theory (MRT)**	Daft and Lengel, 1986	1986	13652	Yadav et al., 2014	2014	537
Marketing	**Trust Commitment Theory (TCT)**	Morgan and Hunt, 1994	1994	33690	Ameen et al. 2021	2021	124
Marketing	**Consumer-Brand Relationship Theory (CBRT)**	Fournier, 1998	1998	11874	Chen et al., 2015	2015	96
Marketing	**Brand Personality Theory (BPT)**	Aaker, 1997	1997	13745	Dickinger and Lalicic, 2016	2016	60
Marketing	**Uses and Gratification Theory (UGT)**	Katz, 1973	1973	3989	McLean and Osei-Frimpong, 2019	2019	249
Information Science	**Technology Acceptance Model (TAM)**	Davis, 1989	1989	69969	Cantallops and Salvi, 2014	2014	1029
Information Technology	**Unified Theory of Acceptance and Use of Technology (UTAUT)**	Venkatesh et al., 2003	2003	40206	McLean and Osei-Frimpong, 2019	2019	249
Information Science	**Diffusion of Innovation Theory (DIT)**	Bass, 1969; Rogers, 1995	1969	141620	Kumar et al., 2016	2016	177

connections and the user's bearings in the network hold importance. An influential user, also deemed a social media Influencer, has a central placement in the network and a substantial number of connections. Brands using social media as a central route to engage with consumers use AI to gauge social networking portfolios, target the right communities and use the right influencers. Such structured plans lead to brands enjoying a sustainable business scenario.

4.2. Social identity theory

The social identity theory was built and presented in the 1970s (Tajfel et al., 1979) to analyse intergroup relations. In the 1980s, the theory was further developed to explain the nuances of social groups and group processes (Abrams & Hogg, 1988; Turner, 1982, 1985; Turner & Reynolds, 1987; Ashforth & Mael, 1989). The social identity theory delves into the nature of self-conception and the corresponding processes (cognitive/affective) and social ideologies in groups and intergroup relationships. Thus, the theory is highlighted as an interactionist social psychological theory that, with time and relevance, gets extended into aspects of group norms, the power of social influence, leadership, deviance within groups, and collective behaviours. Recently, brands used AI techniques to engage, enrich and enamour customers more effectively on social media platforms. Elements of the social identity theory, like social categorisation, social comparison, and social identification, are significant aspects brands should consider while communicating.

The Social Identity Theory (SIT) explains the intricacies of group norms and social impact. Within AI-driven branding, the theory helps explain how consumers identify with brands in the age of advanced futuristic technologies. The emergence of AI has opened the doors to disruptions and inventions. In such a scenario, the SIT can expound on the changes in brand communities and social media platforms which bring about challenges or opportunities for brand sustainability. The impact of AI on the digital world witnesses deindividuation, social mobilization, and possible anomalies within social groups. Symbolic and social implications accompany the prevalence of AI, which the SIT can explain.

4.3. Theory of planned behaviour

The Theory of Planned Behaviour (TPB), developed by Ajzen (1985), is an annexe of The Theory of Reasoned Action (TRA) (Ajzen & Fishbein, 1975). According to the TPB, the variable perceived behavioural control moderates the impact of attitude and subjective norms on behaviour. A person's intention to act or behave a certain way depends on the individual's evaluation of the behaviour (attitude), the perceived social expectations to perform or not perform (subjective norms), and the perception of how easy the task is (perceived behavioural control). The theory has been used to explain the intention behind the behavioural choices of social media users (Casaló et al., 2010; Chang & Zhu, 2011). The TPB has been noted in literature for its parsimony and unique contribution to explaining behaviours (Ajzen, 1991).

The Theory of Planned Behaviour (TPB) helps evaluate the behaviour of social media users. The psychological theory connects intentions to behaviours by focusing on attitude, subjective norms, and perceived behavioural control. AI-driven branding could use the TPB to understand the causes of users' behaviour towards a particular digital campaign, an engaging activity, or brand messaging. Understanding and evaluating the behavioural inspirations of consumers will help brands to build a sustainable strategy with the help of advanced technologies.

4.4. Social cognitive theory

The Social Cognitive Theory characterises human behaviour as a triplex, vibrant, complementary interaction of three elements: the social system/networks, behaviour, and personal factors (Bandura, 1986). The theory proposes that two concepts, self-efficacy, and outcome expectations, immensely impact human activity and are at the heart of the theory. From the lens of AI-driven branding, the judgment of both – the ability to carry out the task and the expected consequence of performing the task is essential for brands to assess. The use of the internet or social media platforms and associated consumer cognitive behaviours can be analysed using the social cognitive theory. The theory has gained traction in the Information Systems literature to explain online behaviours.

The Social Cognitive Theory is another psychological theory that analyses and elucidates human functioning. According to the theory, human behaviour is an outcome of several factors. However, the two factors central to the concept are self-efficacy and outcome expectations. These factors help in understanding the consumer behaviours engendered by the adoption of AI in brand activities. Personal cognition plays a core role in users' responses to advanced AI-enabled branding. Favourable consumer responses to AI-driven brand engagement initiatives will build a strong brand image and sustain positive brand associations.

4.5. Source credibility theory

A well-known theory to explain the factors that make a speaker or presenter (in this case, the brand) a credible source of communication is the Source Credibility Theory (SCT). The theory was built by Hovland et al. (1953). From a theoretical lens, the SCT facilitates understanding of the variables that affect the acceptance of the message by the receiver. The two primary constructs that the theory highlights as responsible for impacting the acceptance of communication are – source expertise (extant knowledge on the subject) and source trustworthiness (honesty, sincerity, and believability) (Hovland et al., 1953). The SCT has been applied in eWOM literature to explain how experts influence online information helpfulness (Filieri, 2015), the acceptance of information communicated by brands (Cheung et al., 2009), and the impact on buying intentions (Filieri et al., 2018; Ismagilova et al., 2021).

The user's first impressions of the brand's social media handle, the website, or the branded content help establish credibility. The Source Credibility Theory helps break down how a brand can ensure the credibility of integrated brand communication via advertisements, digital banners, social media content, and other offline or online engagement strategies. Using AI, brands can predict their target audience's choices and preferences. Thus, the Source Credibility Theory can help in assessing the power of eWOM, as influencers tie up with brands to market the product on social media. The purchase intention of viewers depends on the perception of the credibility of the expertise and trustworthiness of the source.

4.6. Theory of cognitive dissonance

One noted social psychology theory is The Theory of Cognitive Dissonance (TCD), established by Leon Festinger (1957). The TCD is a psychological delineation of a situation when a person experiences two or more cognitions (thoughts) that are contradictory to one another. Such situations result in the individual experiencing an unpleasant feeling arising from the conflict of thoughts. The feeling persists until they can resolve the condition by altering their cognitions (Festinger, 1957). The TCD comprises a four-fold process beginning with Cognitive Discrepancy, followed by Dissonance, Motivation, and Discrepancy Reduction.

The Cognitive Dissonance Theory (CDT) is widely accepted in the era of futuristic AI technologies. The CDT helps evaluate why some brand initiatives involving AI do not go as planned. Often consumers may experience some latent psychological apprehensions when their behaviour/responses are not aligned with their underlying beliefs. This dissonance highlights negative perceptions toward brand initiatives.

AI-enabled activities must consider the related anxieties of using the brand and the attitude change to overcome the dissonance. Both aspects affect purchase intention and buying decisions. The emergence of advanced technologies and its integration into shopping experiences has led to a sense of dissonance among some consumers who are still not ready to adopt or accept the technological intervention. A sense of dissonance can also occur when news spreads of how particular brands are misusing AI tools to invade consumer privacy and other unethical practices. However, AI also helps largely in consistently maintaining loyalty and post-purchase satisfaction (Sakas et al., 2023) through predictive analytics, personalised recommendations (Acharya et al., 2023), and proactive addressing of complaints through chatbots and virtual assistants. Prompt AI technologies can help in reducing consumer dissonance, mitigating consumer vulnerability (an outcome of dissonance) and boosting positive opinions of the brand.

4.7. Para-social relationships theory

The Para-Social Interactions/Relationships Theory introduced by Horton and Richard Wohl (1956) describes the communicative interplay between media characters and their audience. The theory highlights the illusion of intimacy created by the interactions that occur. The audience includes the users of media. The media characters of the brand could consist of an array of personalities like presenters, actors, celebrities, and animated figures, any of whom could interact through an interaction event (a television program, commercial, or talk show interview). The behaviour of the media character and interactions develop a kind of illusion of interpersonal contact between the media figures and the users, which could take place in a single viewing episode (para-social interaction) or beyond a viewing episode (para-social relationship) (Horton & Richard Wohl, 1956). Horton and Richard Wohl (1956) defined Para-Social Interaction as a "simulacrum of conversational give-and-take" (p.215). The theory emphasises how media users perceive a para-social interaction or relationship as a personal and mutual social interaction, even though it is only an illusion.

The Para-social Theory estimates the quality of the interaction between the media platform and the user. There are two facets to the Para-social Theory. A single viewing episode is linked to para-social interactions, and extending a single viewing episode refers to the para-social relationship. With the entry of AI, there are new avenues of

interaction like chatbots, virtual assistants, AI companions, brand avatars, and social media characters. The Para-social Theory considers intimacy, connections, trust, and bonding during such interactions. The media consumed displays a persona that establishes an illusion of an interaction or a relationship with the viewer.

4.8. Consumer culture theory

The Consumer Culture Theory (CCT), constructed by Arnould and Thompson (2005) is an interrelated set of theoretical views that explore the interplay between consumer behaviours, the market, and cultural implications. Despite embodying many distinct theoretical processes and research ambitions, the CCT flows into a standard theoretical orientation. To examine the cultural complexities in the marketing ecosystem, the CCT does not explore culture as a harmonious and smooth collection of shared descriptions, ideologies of life, and unified values by members of society (Arnould & Thompson, 2005). Instead, the CCT jumps into the heterogeneity and multiplicity of cultural groups in the broader scheme of things like globalisation and market capitalism (Arnould & Thompson, 2005).

The Consumer Culture Theory (CCT) is a relatively recent theory that builds on how consumers confidently create their narratives by revising branded content, advertising messages, and other mass-media communications. The CCT shines a light on how consumers are taking charge in reworking and meditating on integrated brand communications such that narratives resemble the existing culture, social influences, and symbolic meanings.

AI-driven branding can use the takeaways from the CCT to build a more effective mode of storytelling, such that consumers can relate to brand activities when the varied types of cultures are considered and represented. The CCT brings in cultural dimensions that explain consumer behaviour. Thus, brands can capitalise on the learnings from the CCT to align and strengthen their brand identity by looking into the cultural values of consumers and the ethical ramifications of AI implementation. As AI is dynamically transforming the way consumers and brands interact the CCT is a theory that will influence consumer-brand relationhsips, personalisation, consumer power positions, brand narratives and the creation of consumer-brand cultural communities.

4.9. Media richness theory

Daft and Lengel established the Media Richness Theory (MRT) in 1986 as a literary extension of the Information Processing Theory. The MRT is also referred to as the Information Richness Theory. The theory helps us understand how a communication channel or medium can replicate the information shared over it previously. According to the MRT, different channels of media vary in richness. The term "richness" is defined in literature as "the ability of information to change understanding within a time interval" (Daft & Lengel, 1986, p. 560). The theory can help decipher the richness of social media channels and how AI can augment or enrich communication with consumers. AI can also help assess the reaction or response to such channels and the corresponding content to determine the richness of the medium.

In the age of AI, branding can use the insights of the Media Richness Theory to propagate the right messages via the right media channel. With the advent of new AI-led platforms, the brand should pay close attention to sharing a relevant message on the platform that promises adequate or more richness such that the message is successfully delivered and understood.

4.10. Trust commitment theory of relationship marketing

According to Morgan and Hunt (1994), trust and commitment to a relationship play a colossal role in building and sustaining relationships between buyers and sellers. Thus, the core of the Trust-Commitment Theory of Relationship Marketing was established by Morgan and Hunt in 1994. The theory is applied in an assortment of contexts like collective buying websites (Wang et al., 2016), brand relationships in online communities (Zhang et al., 2018a), social media fan pages (Akrout & Nagy, 2018), and online retailing (Elbeltagi & Agag, 2016). Trust is vital in increasing relationship commitment between customers and retail brands on online platforms like social media (Wang et al., 2019a). Trust and commitment emerge as crucial in AI-mediated associations, interactions, and conversations between brands and consumers. From the perspective of AI automation, technology-driven platforms, and AI conversational agents, trust is an elementary concern and a solid reason for hesitancy to accept AI methods. Concerns like data vulnerabilities, data privacy, control of data sharing, and ethical loopholes need to be addressed with the help of insights from the theory.

The Trust-Commitment Theory stresses the parts trust and commitment play in the relationship outcomes between consumers and brands. Long-lasting relationships developed between buyers and sellers depend on the portions of trust and commitment that exist between the two parties involved. In AI-driven branding, the theory is applied to relationships in online brand communities, social media interactions, and associations with AI companions (chatbots, virtual assistants, and robots). Both trust and commitment are elementary factors that comprise a healthy relationship in a technologically fronted brand world. Automation and predictive capabilities enable the progress of brand quality. However, at the same time, such aspects deter ethical prospects like privacy, consumer vulnerability, and data breaching. The Trust-Commitment Theory helps brands use AI to consider the power of incorporating elements that will enable the brand to sustain and endure market challenges.

4.11. Uses and gratification theory

The Uses and Gratification Theory (UGT) is a theoretical motivational paradigm (Katz, 1959) wherein it is suggested that since users are aspirational and determined to fulfil their needs, they select media that matches and gratifies their needs (Katz et al., 1973). The UGT is a theory of mass communication deeply rooted in communication science to gauge why consumers pursue and opt for a specific choice of media to quench their media consumption needs (Eighmey & McCord, 1998; Gallego et al., 2016). Inspecting the factors influencing media choices and how they satisfy consumer needs has formed a significant part of social media research (Chen, 2011; Dholakia et al., 2004; Porter & Donthu, 2008). The UGT theory considers social and psychological elements of needs (Wurff, 2011). It is used to explore the drivers that lead to technology adoption (Grellhesl & Punyanunt-Carter, 2012) as the scope of the theory is extensive (Luo & Remus, 2014).

Technology has been and continues to be a game-changer in branding. However, understanding the driving factors that encourage the adoption of AI in brand activities can be done by considering the Uses and Gratification Theory (UGT). The UGT paradigm contains insights into the satiation of needs and desires that motivate actions. The theory is used in the AI-driven branding domain to decipher the motivation for consumers to interact with brands on websites, social media pages, brand community sites, online gaming portals, and virtual/augmented reality platforms.

4.12. Consumer-brand relationship theory

The Theory of Consumer-Brand Relationships highlights the nuances of consumers' interpersonal emotional, and cognitive relationships with brands (Fournier, 1998; Veloutsou, 2007). The variables that feature in the framework and determine the strength or nature of brand relationships are love, self-connection, commitment, interdependence, intimacy, and brand partner quality.

The Consumer-Brand Relationship (CBR) Theory emerges from the marketing domain to deem a brand a legitimate relationship partner. AI technology-mediated branding can delight consumers with sustained

engagement. Thus, generating positive, in-depth emotions like love, intimacy, commitment, and strong connections. The CBR Theory can be helpful to brand managers as the insights can help them create a strong yet personalised brand campaign that will promote sustainable business propositions.

4.13. Brand personality theory

The Brand Personality Theory elucidates a set of human characteristics attributed to a brand (Aaker, 1997). The essence of anthropomorphism is the theory's core, as humane personality attributes describe a brand's persona. The theory aims to understand how consumers identify with the brand by focusing on five broad personality traits: sophistication, excitement, competence, sincerity, and ruggedness (Aaker, 1997).

The Brand Personality Theory rides on the theme of anthropomorphism. Thus, the theory perfectly fits the context of AI-driven branding. The theory's core is associating human personality attributes with the brand and visualising the chances of congruence with a consumer's personality. AI in branding brings AI companions and other avenues of brand interaction. Such interactions allow the brand to hone and shape its personality to facilitate better identification and connections with the target audiences. The deeper the connections, the stronger the sustainable scaffolding that protects the brand in the market.

4.14. Technology acceptance model

The Technology Acceptance Model (TAM), developed by Davis (1989), has been extensively used for assessing the acceptance, adoption, and use of emerging technologies (Davis et al., 1989). The TAM is one of the dominant and leading research frameworks in information technology to examine the determinants predicting the intention and acceptance to use technologies. The two primary concepts that the theory introduces are perceived usefulness (the belief that using the technology will improve performance) and perceived ease of use (the belief that the use of the technology is simple and easy) (Davis, 1989). Both variables help envisage the users' evaluation of using the technological application. The conceptualisation and justification of TAM should be enriched by integrating other specific emerging technology-related constructs like AI techniques (Legris et al., 2003; Serenko et al., 2008).

The Technology Acceptance Model (TAM) orbits around forecasting the chances of acceptance and use of emerging technologies. AI is a technologically growing capability to reckon with, and most brands are still learning the art of imbibing AI tools for better reach, targeted predictions and sustainable business models. The inherent insights from TAM can help examine the present adoption of technology and the potential for future adoption.

4.15. Unified theory of acceptance and use of technology

The Unified Theory of Acceptance and Use of Technology (UTAUT) is a unified theoretical viewpoint merging eight established theories (Venkatesh et al., 2003). The UTAUT contains four critical determinants of intention and usage and four moderators that complete the model. The selected eight theories embedded in the UTAUT include the theory of reasoned action, the technology acceptance model, the motivational model, a model combining the technology acceptance model and the theory of planned behaviour, the theory of planned behaviour, the model of PC utilisation, the innovation diffusion theory, and the social cognitive theory.

The Unified Theory of Acceptance and Use of Technology (UTAUT) is a fine amalgamation of eight established theories/models on the acceptance and use of technology. The UTAUT provides a holistic perspective on adopting, accepting, and using AI technologies in branding. The unified understanding provided by the theory and associated constructs enables the creation of sustainable brand businesses. The factors that drive the acceptance and the corresponding outcomes are elucidated within the unified theoretical viewpoint.

4.16. Diffusion of innovation theory

The Diffusion of Innovation Theory (DIT) focuses on understanding the process, the reasons, and the speed of the movement of new ideas and technology (Rogers & Cartano, 1962). Everett Rogers introduced the DIT in 1962, which was further fine-tuned by Rogers in 1995. The pace at which innovative ideas and emerging technologies proliferate in the marketplace is the core of the DIT. The theoretical spirit of the model focuses not on convincing people to change. Rather, it is the brand's progression or "reinvention" to become more suited to the needs of the individuals and groups. The innovation is dynamic, not the people (Rogers & Cartano, 1962; Les Robinson, 2009). Diffusion transmits innovations through specific social system platforms (Rogers & Singhal, 2003).

AI-driven branding brings enormous potential for sustainable innovation, ideas, and inventions. The Diffusion of Innovation Theory can help brand managers evaluate the manner and momentum with which contemporary yet futuristic innovations reach the different categories of consumers.

5. Towards a theoretical framework

All constructs have been identified as drivers, moderators, mediators and outcomes. Furthermore, the constructs have been categorised under three buckets basis the discipline the construct emerges from. The thematic analysis of prominent theories underpinning AI-Driven branding unfolds the interplay between the drivers, mediators, moderators, and outcome variables. Fig. 2 offers a birds-eye view of all the theoretical constructs used to expound emerging AI technologies' impact on branding.

5.1. Drivers of AI-driven branding

The first set of drivers describing AI for branding comes from the social psychology discipline. *Subjective Norms* rooted in social psychology refer to the apparent social pressure to execute the behaviour (Fishbein & Ajzen, 1975; Ajzen, 1991). Subjective norm was the core construct used in the Theory of Reasoned Action (TRA) and Theory of Planned Behaviour (TPB). The *attitude is* significant in bringing the evaluative quotient into the equation. Both subjective norms and attitudes affect the perspective and the decision to accept AI tools for branding. The concept of *Perceived Behavioural Control* (PBC) (Ajzen, 1991) is an important driver of the intention to adopt emerging technology like AI innovations by brands (Cantallops & Salvi, 2014). PBC assesses whether the task will be easy or difficult, affecting the desire to try and use modern technology. The concept that differentiates the TRA from the TPB is the PBC. *Self-Efficacy* from Social Cognitive Theory (Bandura, 1986) complements PBC in adopting AI for branding. Four strategic self-efficacy interventions: performance accomplishments, vicarious experience, verbal persuasion, and emotional arousal (Bandura, 1986), compel brands to consider AI for the transformation of brands. The TPB comprising subjective norms, attitude, and PBC, renders a rationale for participation and behavioural intention to adopt AI tools (Cantallops & Salvi, 2014).

Socio-Cultural Factors and *Outcome Expectations* from the Social Cognitive Theory (Bandura, 1986) act as facilitators or impediments to AI adoption in branding. The outcome expectations involve predicting the consequences of accepting AI techniques in branding. More specifically, the concept revolves around personal expectations of these consequences, including esteem and accomplishment (Compeau & Higgins, 1995). *Reciprocal Determinism* considers three parties, the person (with some learned experiences in AI-infused technologies), the environment

(in a social context), the behaviour (responses to accept and use the AI tools), and the vigorously reciprocal interaction between all three. *Observational Learning* refers to how consumers can observe another consumer's behaviour and successfully demonstrate that AI-driven branding is a unique experience, thus believing that even they can perform and complete the task effectively (Bandura, 1986).

Free Choice and *Behavioural Commitment,* from Cognitive Dissonance Theory, instigate consumers to interact with chatbots, engage with brand virtual assistants, create an avatar on virtual reality applications, or create content on a brand social media channel (Colicev et al., 2018; Festinger, 1957). *Perceived Attractiveness* from Para-social Interaction Theory (Horton & Richard Wohl, 1956) inspires consumers to interact with brand-related AI avatars, assistants, or bots. The interaction creates an illusion of intimacy during interactions and fosters a relationship that carries over to the next interaction (Horton & Richard Wohl, 1956). The illusory interpersonal relationship determines the usage level of AI and the corresponding impact on branding (Pitardi & Marriott, 2021).

The second theoretical lens used to define drivers of AI-Driven branding comes from marketing theories. *Shared Value* comes from the Trust-Commitment Theory of Relationship Marketing (Morgan & Hunt, 1994) and represents consumers' common beliefs regarding brand values. The positive shared values between the two parties will steer AI-driven branding onwards and upwards. The next precursor that helps explain the relationship between AI and branding is the dimensions of the *Consumer Culture Theory* (Arnould & Thompson, 2005), which include the following – *Consumer Identity Project, Marketplace Cultures, Socio-historic Patterns of Consumption,* and *Mass-mediated Marketplace Ideologies*. Consumer Culture Theory explores the meanings of culture in the broader social frame of reference to globalisation and technological advancements. More importantly, the constructs used in this theory underscore the socio-cultural dynamics that navigate the AI consumption cycle, resulting in the advancement of theoretical dialogue. *Brand Personality* represents the human attributes associated with a brand (Aaker, 1997). Specific brand personality dimensions like Competence or Sophistication can engender consumers' support and acceptance of AI-induced branding efforts (Dickinger & Lalicic, 2016). Factors like *Media expectations* highlighted in the Uses and Gratification Theory (Katz, 1973) strongly motivate AI-driven branding to be successfully implemented and accepted. There must be a sense of need and expectation to use AI techniques while interacting with the brand.

The third and final category takes its inspiration from the field of Information Sciences. *Perceived Ease of Use* and *Perceived Usefulness* are two concepts derived from the Technology Acceptance Model (TAM) (Davis, 1989). These two concepts have high prediction capabilities when envisaging the intensity of acceptance of new and advanced technologies like AI, machine learning, text analytics, and many more such techniques (Cantallops & Salvi, 2014). The *Facilitating Conditions*, a construct taken from the UTAUT (Venkatesh et al., 2003), is a factor to consider as the availability of the right conditions conducive to using the AI tools and mechanisms precedes the question of whether it is to be accepted by the user (McLean et al., 2019).

5.2. Moderators

Underpinning theories from the Social Psychology domain hinted at the important role of strong ties, source expertise, and source trustworthiness as moderators. *The strength of Ties* from the Social Network Theory (Granovetter, 1973) indicates the role of network ties potency in determining the influence of AI on branding. Besides the strength of ties, the other two moderators, *Source Expertise* and *Source Trustworthiness* from the Source Credibility Theory (Hovland et al., 1953), ascertain the acceptance of AI for brand transformation.

Marketing theories propose relationship cost and relationship benefits as important moderators to ascertain the influential role of AI-Driven branding. Both relationship costs and benefits are rooted in the Trust-Relationship Commitment Theory of Relationship Marketing (Morgan & Hunt, 1994). The Relationship costs refer to the "switching costs" incurred when terminating the ongoing connection for an alternative one. The relationship benefits refer to the superior and unique features a brand offers its customers to garner a competitive edge.

The third school of thought pertains to information science theories. *Effort Expectancy* and *Performance Expectancy* rooted in UTAUT have a moderating effect on the dynamic brand relational model. Another underrated moderator in the relational equation is *Time*. The time required to execute an AI-oriented project, to create awareness of a new AI initiative, and the response time of AI-Driven branding have a moderating effect.

5.3. Mediators

Underpinning theories from the Social Psychology domain proposes the important mediating role of *Para-social interactions* (Horton & Richard Wohl, 1956) between consumers and brand media platforms, characters/avatars, or brand ambassadors. The other intervening variables come from Branding theories, which collectively highlight the importance of *Communication*. Elements of the Media Richness Theory (Daft & Lengel, 1986) include media elements like *Face-to-Face Communication, Videoconferencing, Telephone, Written Addressed Documents,* and *Written Unaddressed Documents. Social Systems* and *Communication Systems* broadly assist in explaining how AI impacts branding, as these communication channels either act as a boon or a bane in terms of usage information science theories' perspective points at the mediating role of *Innovation* in enticing consumer interest for AI-Driven branding.

5.4. Outcomes

AI-driven branding has numerous outcomes from social psychology, marketing, and information science perspectives. From a Social Psychology perspective, social categorisation, social comparison, and social identification emanating from social identity theory (Tajfel et al., 1979) are important outcomes of AI-driven branding. AI technologies (like sentiment analysis, text analytics, and machine learning algorithms) concatenated with various social media platforms/digital media channels help in deriving insights basis the brand-related user-generated content that shows how consumers/brands are socially categorised, compared, and identified. In addition to social identity, forced compliance rooted in cognitive dissonance theory compels consumers to alter their conflicting cognitions by accepting AI-driven branding (Festinger, 1957; Colicev et al., 2018).

The next set of outcomes is rooted in marketing theories. One desired outcome is building and nurturing consumer-brand *relationships* (Fournier, 1998). Consumer brand relationship encapsulates *Love, Self-Connection, Commitment, Interdependence, Intimacy*, and *Brand Partner Quality*. Brands expect to build consumer relationships with AI techniques by generating deep insights about customer needs and preferences. Brand Trust is another vital outcome as it negates doubts like data privacy and other ethical ramifications. Trust is inspired by the Trust-Relationship Commitment Theory developed by Morgan and Hunt (1994). *Need Gratification* taken from the *Uses and Gratification Theory* (Katz, 1973), results in the desired customer experience when interacting, consuming, and reminiscing AI-driven branding.

The third and final set of outcomes, such as *Acceptance* and *Intention to Use* embodied in Information Science theories. Theories like TAM and UTAUT (Davies, 1989; Venkatesh et al., 2003) propose behavioural intention as a precursor to AI-driven brand acceptance. Most brands contemplating introducing AI-enabled branding aim to accurately assess consumer acceptance chances (Cantallops & Salvi, 2014; McLean et al., 2018). Consumers' acceptance and positive behavioural intention to use AI-enabled brands continue to be the desired outcome.

5.5. Research propositions

5.5.1. Attitude, choice of communication, and brand partner quality

Favourable attitudes towards AI technology influence the quality of brand partnerships and this relationship. The choice of communication can pose as a mediator in the relationship. In a study on consumer attitudes towards AI, researchers found that consumers with favourable attitudes towards AI were likelier to trust the brand and showcase true purchase intentions towards products that incorporated AI technology (Li et al., 2020). This suggests that positive attitudes toward AI can positively influence brand partnerships.

A study by Cheng and Jiang (2021) highlights the components of Chatbots Marketing Efforts (CME) components: interaction, information, accessibility, entertainment, and customisation. CME has a direct impact on the quality of communication with chatbots and an indirect effect on consumer-brand relationships. Nguyen et al. (2021) examined the influence of AI on customer-brand identification, customer advocacy, and customer experience flow.

A study by Veloutsou (2015) depicted that the strength of consumer-brand relationships predicts brand loyalty. Consumer-brand relationships do not play a moderating role. Instead, they mediate the relationship between brand trust, loyalty, and satisfaction.

Together, the extant literature suggests that an affirmative attitude toward AI can influence the quality of brand partnerships. When examined with mediation by choice of communication, the brand-partner relationship will decipher how brands that incorporate AI technology in their communication strategies can leverage and enhance their businesses.

RP1: Favourable attitudes toward AI technology influence brand partner quality mediated by choice of communication.

5.5.2. Perceived attractiveness, performance expectancy, and acceptance of AI technology

In a study on consumer acceptance of AI-powered chatbots, it was observed that they perceived the chatbots as This suggests that performance expectations influence perceived attractiveness Maroufkhani et al. (2022) draw on the Uses and Gratification Theory, Signalling Theory and Prospect Theory to decipher that perceived privacy risk is a critical factor affecting the consumer's holistic perspective towards voice assistants. Brand loyalty of consumers develops with the consistent use of voice assistants.

The study by Fernandes and Oliveira (2021) divulges how social, functional, and relational elements engender technology adoption, disentangle crossover complexities, and unfurl the moderating role of experience and the need for human interaction. Another study examined the role of effort in consumer acceptance of AI-powered virtual assistants/chatbots for healthcare use and found that consumers were more likely to accept the technology when it needed low effort, which resulted in high performance (Jiang et al., 2022). The analysis suggests that effort and performance expectations must be considered when assessing the relationship between perceived attractiveness and consumer acceptance. Cumulatively, the studies indicate that the perceived attractiveness of emerging AI technologies can increase consumer acceptance to use them in branding, and performance expectations can moderate this relationship.

RP2: Perceived attractiveness increases the acceptance of emerging AI technologies in branding when moderated by the performance expectations of the consumer.

5.5.3. Perceived attractiveness, source trustworthiness, and acceptance of AI technology

When a study analysed consumer acceptance of AI-powered chatbots, it was found that the source trustworthiness of the provider had a significant impact on consumers' perceived attractiveness of the chatbot (Ameen et al., 2022). Consumers were more likely to find the chatbot attractive when a reputable and credible source, such as a well-known brand or a trusted institution, provided it. A study by Masuda et al. (2022) used the Theory of Persuasion to create a model considering three personal elements (physical attractiveness, attitude homophily, and social attractiveness) and specific characterisations, including trustworthiness and perceived expertise.

Furthermore, to gauge the power of information processing on the blog, the study by Chu and Kamal (2008) examines the impact of perceived blogger trustworthiness on blog readers' assessment of brand messages and the effect of the argument quality. The study reveals how perceived blogger trustworthiness substantially impacts forming brand attitudes. The study by Yang et al. (2021) highlights the favourable effect of perceived source genuineness brand authenticity with source trustworthiness mediating the interaction.

Brands that incorporate AI technology in their branding strategies should consider the reputation and credibility of the source providing the technology to enhance its perceived attractiveness and increase consumer acceptance. Consumers are more certain to accept AI technology when a reputable and trustworthy source provides it with expertise in the relevant field.

RP3: The influence of perceived attractiveness on acceptance of AI technologies in branding will be moderated by source trustworthiness.

5.5.4. Subjective norms, time, and brand love

A study on consumer attitudes toward AI-powered chatbots highlights how social norms significantly influenced consumers' emotional involvement in using a chatbot (Giroux et al., 2019). In the short term, social norms have a weaker impact on love and usage, but over time, consumers may become more influenced by their social environment, and social norms have a more substantial effect. Wang et al. (2019b) looked at the impact of subjective norms on brand love and found that while subjective norms had a significant positive impact on brand love, the effect was stronger over time as consumers became more attached to the brand. Brands adopting AI technologies in their branding must assess the socio-cultural factors and be aware of the role such factors play in influencing brand love.

RP4: The impact of subjective norms on brand love when brands use AI technology is moderated by time such that the effect is stronger in the long term.

5.5.5. Socio-cultural factors, time, and brand love

In a study by Verma (2021), a holistic understanding of the drivers of equity is presented for awareness of the conceptual insights on the enablers of brand equity formation. As brand love and brand equity mediate the relationships, socio-demographic aspects are also considered. Another study examined the role of cultural factors in consumer acceptance of AI-powered products and services and found that cultural factors such as individualism and collectivism had a significant impact on consumer attitudes, but the effect was stronger in the long term (Wu et al., 2021). Over time, consumers' cultural values became more ingrained and strongly influenced their attitudes toward AI-powered products and services.

Interestingly, Hollebeek and Belk (2021) introduced the Consumers Technology-Facilitated Brand Engagement (CTFBE) concept and aligned it with the CCT. The following aspects were investigated: consumption myths, rituals, brand communities, consumption, and identity issues.

Brands that use AI technology in their branding strategies should consider the long-term effects of socio-cultural factors and work to build strong, positive associations with their brand over time to enhance brand love.

RP5: The impact of sociocultural factors on brand love when brands use AI technology is moderated by time such that the effect is stronger in the long term.

5.5.6. Media expectations, para-social relationships, and brand trust

The literature comprises a study by Tatar and Eren-Erdoğmuş (2016) that reveals how clear websites, website security, effective online interactivity, and collaborations with other established websites positively impact brand trust.

Furthermore, a study by Akrout and Nagy (2018) highlights the importance of relationship quality to develop trust within brand fan pages. The trust transcends into positive word-of-mouth (WOM) for the fans on the page. The study reveals that demographics play a crucial role as young female fans display high levels of engagement, build a strong relationship with the brand, and thus spread positive WOM.

RP6: The influence of media expectations on brand trust is mediated by para-social relationships.

5.5.7. Consumer culture, para-social interactions, and brand love

Research suggests a positive brand culture can influence Brand love. Positive brand culture is reflected by the shared values, beliefs, and behaviours of the organisation and its employees, which are aligned with the consumer's values. A favourable brand culture can foster positive emotions and attitudes towards the brand, leading to brand love.

Para-social relationships refer to people's psychological connections with media characters, such as celebrities or fictional characters. These relationships can influence people's perceptions of and attitudes toward brands. Understanding how such interactions can mediate the effects of needs and media expectations on brand trust and commitment is important.

For example, a study by Ahn et al. (2019) found that para-social interactions with brand representatives mediated the relationship between brand culture and brand love. The study found that a positive brand culture led to a greater sense of identification with the brand representative, leading to greater brand love.

A study by Kim et al. (2021) speaks of the positive impact of interdependent self-construal on brand loyalty. The relationship is mediated sequentially by Para-social interactions and consumer engagement. Furthermore, Junaid et al. (2019) indicate how hedonic pleasure and escapism directly affect brand love. On the other hand, flow, challenge, and learning indirectly impact brand love. The study highlights the way the relationship between consumption experience and customer engagement is mediated by brand love.

Correspondingly, a study by Shin et al. (2019) found that para-social interactions with social media influencers mediated the relationship between brand culture and brand love. The study found that a positive brand culture, which focused on social responsibility and sustainability, led to a greater sense of identification with the social media influencer, leading to greater brand acceptance. Finally, in the study by Tsai et al. (2021), we see how perceived para-social interactions and dialogue mediate the relationship between chatbot social presence communications and consumer engagement. The chatbot's human-like traits and design ignite favourable effects of social presence messaging through psychological mediation.

Thus, it will be interesting to see how positive brand culture can foster positive emotions and attitudes towards the brand, and para-social interactions reinforce these emotions and attitudes by creating identity congruence and connection with the brand representative or social media influencer.

RP7: Brand love is the outcome of consumer culture dynamics when mediated by para-social interactions.

6. Discussion

The constructs embedded in the integrated theoretical framework, together build a holistic understanding of AI-driven branding. The integrated theoretical framework stands on the broad shoulders of the metaphorical giants of scholarly literature. Every chosen theory lends its explanation prowess as the associated constructs are heavy-laden with rich meaning and rationale. The integrated theoretical framework presents triadic implications- theoretical, practical and policy implications. Considering the plethora of theories used to explain varied facets of AI-driven branding, the focus has tapered and resulted in fragmented literature. The integrative nature of the present review curates significant brand research outcomes. The integrated theoretical framework makes novel contributions to emerging literature, spawns strategically creative ideas for practitioners, and highlights policy concerns. The present study aims to create a robust understanding of how to use AI mechanisms to build a sustainable brand environment that promises trust, positive perceptions and customer delight. Table 3 elucidates a summarised depiction of the seven research propositions developed in the present analysis. Each of the research propositions highlighted in the study aim to have a catalytic impact on future AI-driven brand research. The implications are vital as practitioners and academicians view AI as the future of brand engagement, relationships, and success (Verma, 2021).

The integrated theoretical framework is created with the objective of keeping up with the rapid and consistent advancement of AI technologies. In order to ensure that the model is able to sustain the speedy and constant evolution of AI, we have attempted to build a structure that is flexible, holistic and transparent enough to endure the strong

Table 3
Summary of the research propositions.

Constructs	Research Proposition	Research Articles
Attitude; Choice of Communication; Brand Partner Quality	*Favourable attitudes toward AI technology influence brand partner quality mediated by choice of communication.*	Li et al. (2020); Cheng & Jiang (2021); Nguyen et al. (2021)
Perceived Attractiveness; Performance Expectancy; Acceptance of AI Technology	*Perceived attractiveness increases the acceptance of emerging AI technologies in branding when moderated by the performance expectations of the consumer.*	Jiang et al. (2022); Maroufkhani et al. (2022); Fernandes and Oliveira (2021)
Perceived Attractiveness; Source Trustworthiness; Acceptance of AI Technology	*The influence of perceived attractiveness on acceptance of AI technologies in branding will be moderated by source trustworthiness.*	Ameen et al. (2022); Masuda et al. (2022); Yang et al. (2021)
Subjective Norms; Time; Brand Love	*The impact of subjective norms on brand love, when brands use AI technology, is moderated by time such that the effect is stronger in the long term.*	Giroux et al. (2019); Wang et al. (2019a)
Socio-Cultural Factors; Time; Brand Love	*The impact of sociocultural factors on brand love, when brands use AI technology, is moderated by time such that the effect is stronger in the long term.*	Verma (2021); Wu et al. (2021); Hollebeek and Belk (2021)
Media Expectations; Para-Social Relationships; Brand Trust	*The influence of media expectations on brand trust is mediated by para-social relationships.*	Tatar and Eren-Erdoğmuş (2016); Akrout and Nagy (2018)
Consumer Culture; Para-Social Interactions; Brand Love	*Brand love is the outcome of consumer culture dynamics when mediated by para-social interactions.*	Kim et al. (2021); Junaid et al. (2019); Tsai et al. (2021)

technological headwinds. The multi-disciplinary approach and the malleability of the analysis will help future researchers to scale and modify the integrated theoretical framework as new technologies materialise.

6.1. Implications for theory

The present study widens the horizon of the AI-driven branding research domain. The conceptualisation of the integrated framework with the prominent and relevant theories in the extant literature engenders future research with a unified perspective. The theoretical framework highlights the holistic view of the underpinning theories explaining the AI in branding phenomenon. The integrated framework canvas 16 theories rooted in disciplines like psychology, information technology, and marketing. The three disciplinary buckets reflect the theoretical variety and flavour of the domain. The nuanced flavour of each discipline seeps into the integrated theoretical framework, thus, enhancing the integrated interpretation of the relationships represented in the framework. The intricate integration of dominant theories from deeply rooted disciplines led to the understanding that there is scope for overlap of concepts and conceptual conflicts. The potential overlaps led us to believe that the need for a holistic theory is the need of the hour. Each theory that is selected to help explain a certain aspect of AI-driven branding emerges from an established discipline.

Past literature points to the relational aspects of existing theories contextually with the specificity of stakeholders. Thus, limiting the applicability and lacking generalisability. The holistic snapshot of the integrated framework will give future researchers a well-rounded approach to the domain (Lunde, 2018; Torraco, 2005). With the integrated framework, scholars may explore varied research questions, test the interplay of various constructs and assess the avenues for building sustainable businesses for advancing the AI-driven branding domain. Assessing the results of the interplay could lead to new paths, richer research discoveries, and improved logical justifications of findings (Venkatesh et al., 2003) in AI-driven branding research.

With time, scholarly literature in marketing has progressed from using theories with a universal perspective to using theories with a micro perspective (Lunde, 2018). The shift in focus to a micro perspective on the inter-relationship of AI and branding has led to the use of multiple theories to explain the same phenomenon. The integrated theoretical framework presents a macroscopic and holistic view of AI in branding to enable brands in the age of emerging technologies. The integrated theoretical framework resolves the dichotomy of multiple theoretical lenses and offers a rationale for the selection of an explainable theory (Eiriz & Wilson, 2006). Hence the integrated theoretical framework settles the state of dissonance by presenting a holistic representation of all the significant theories and their constructs in a framework to explain the AI-driven branding phenomenon (as seen in Table 3).

6.2. Implications for practice

The integrated theoretical framework impacts marketing practitioners by enlightening them on emerging trends that lead to the evolution of branding using AI machinery. The integrated theoretical framework creates a sound foundation for crafting and endorsing AI-enabled brand strategies and capturing consumers' reactions to implemented strategies. Big data analytics is one example of AI techniques that improve a brand's predictive capabilities leading to strategic differentiation and sustainable competitive advantages. However, most firms are still at the nascent stages of adoption due to the lack of understanding and confidence in the technology (Kwon et al., 2014). Branding and Marketing practitioners need to stay aware of the benefits of AI mechanisms like Big Data Analytics, Natural Language Processing (NLP) and Sentiment Analytics to uncover the power to predict consumer choices (Kushwaha et al., 2021). The integrated theoretical framework facilitates the assessment of the past success of an AI-driven brand strategy, the decision to use AI algorithms in brand activities or the measurement of customer responses to AI capabilities fronted by the brand (Davenport et al., 2019). The holistic framework will make brands aware of the predictor variables and outcome variables that practitioners should consider while formulating and implementing advertising or other brand initiatives.

The integrated theoretical framework is a practical collective intelligence resource (Wilson & Daugherty, 2018) for marketing practitioners. The bank of dominant theories schematically amalgamated into the integrated framework acts as an intellectual source of theories, constructs, and insights emerging from the tried and tested literature. Thus, corporate brand managers and consultants will benefit from a scholastic reference guide in understanding the factors that drive AI-driven branding and the associated outcomes. Brand managers can use the insights emerging from the framework to study, scan and scrutinise their branding plans for crafting sustainable plans to survive in the competitive marketplace. The inspection of brand engagement blueprints will illuminate the flaws and highlight the strengths. Thus, the practitioner is better armed to develop a systematic approach while implementing AI.

Marketing practitioners can leverage the unified viewpoint by screening the driving variables incorporated into the model and applying them to their brand strategy. The variable "Trust" currently is the desired brand outcome and concern to most consumers (Davenport et al., 2020; Martin et al., 2017; Martin & Murphy, 2017). Brand representatives can get a sense of direction from the underlying theoretical insights to gauge the dimensionality propagating AI in branding. Practitioners can strategize on how to make the constructs actionable and manifest the relevant ones into practicality. While considering the construct "brand trust", practitioners can assess the actionable elements of trust. The elements can be translated into a strategic brand plan for practical usage that will have positive implications for both, the brand and the organisation. The integrated theoretical framework raises concerns that practitioners often ask and wish to resolve. The factors that bring to light a sense of risk and lack of sustainability in AI-driven branding include data privacy issues, ethical drawbacks, consumer data vulnerabilities, deep fakes, and users' discomfort with interacting with a bot (chatbot/robot) (Mori, 1970; Davenport et al., 2021). The holistic framework acts as a guiding map enabling marketing practitioners to navigate the deep waters of AI-driven branding and effectively deal with privacy issues, consumer fears and ethical ramifications while on the journey towards brand success.

6.3. Implications for policy

The integrated theoretical framework fosters implications for policymakers by shedding light on elements like trust, privacy, data breach, environmental issues, copyright infringement, and deep fakes. The policy implications encourage brands to focus on the implementation and corresponding integration of sustainable AI-driven brand practices for the creation of a sustainable business model. There is an increased opportunity for policymakers to question the repercussions of AI-enabled brand activities on sustainable marketing practices. Policymakers need to set up ethical guardrails that consistently check on whether there is responsible messaging, environmental consciousness, privacy, and other ethical ramifications (Dellemijm, 2023). The integrated theoretical framework creates an opportunity for changemakers to collaborate with brand managers to co-create or re-create rules and policies for the responsible advancement of AI-driven branding. The rules and policies must address the ethical aspects and propagate trust (Hutchinson, 2022).

Policymakers can encourage the brand to function responsibly and encourage consumption in moderation. Customers should feel gratified and safe while interacting with their favourite brands knowing that the AI technologies used are only to augment the experience. However,

consumer-brand interactions should not occur at the cost of privacy, data breaches, or other unfavourable actions (Davenport et al., 2020; Martin et al., 2017; Martin & Murphy, 2017). Changemakers can use the insights from the integrated theoretical framework to create effective training programs and regular discussions with marketing professionals.

7. Conclusion

AI-driven branding is securing the attention of scholars and practitioners for the agility and precision with which branding initiatives are being implemented to sustain the business in the long-term. AI-enabled technologies, platforms and agents are channelling metamorphic changes in consumer-brand interactions. Thus, the influence of AI in branding is gaining practical attention and piqued researchers' curiosity in recent times (West et al., 2018). Although the AI in branding research domain is growing, comprehensive literature reviews are scant. Among the previous studies, a thematically integrated theoretical framework was missing. The research domain of AI-driven branding requires a holistic theoretical perspective that considers all prominent theories equally. Past researchers have used varied theories basis the context and specific research questions. However, researchers would be caught in a conflict of theoretical choices. Ergo, the present study builds an integrated theoretical framework with the objective of schematically amalgamating the prominent theories that past literature has applied to explain AI-driven branding.

The integrated theoretical framework is a cohesive theoretical representation that emerges from the dependable foundations of the prominent theories used in AI-driven branding literature. The present integrative study analyses 414 articles published in top-indexed journals on AI-driven branding spanning 22 years (2000 – 2022). The study aimed to draw thematic insights from the underpinned theories to construct the integrated theoretical framework. The extensive scientific process with systematic filtering criteria (prominence of use, citations, recency) rendered distinct theoretical perspectives to build a unified framework for AI-driven branding.

The concept of unification was inspired and executed by "abstractly understanding a situation or problem by identifying patterns or connections and key underlying properties" (MacInnis, 2011, p.140). The theoretical framework is developed to be a catalyst for the creation of new conversations (Alvesson & Sandberg, 2020) and re-imagination of the literature to advance the domain of AI in branding. The integrated theoretical framework showcases a precise list of drivers, moderators, mediators and outcomes that cumulatively explain the AI-driven branding phenomenon. Each theoretical construct is extensively explained and accurately analysed. To further explain the intricate relationships between the theoretical constructs and ensure robust theoretical development, research propositions are elucidated. Among the series of propositions, the relationship between attitude and brand partnership quality is explored by considering the mediation of choice of communication. Furthermore, the interplay between sociocultural factors and brand love moderated by time is analysed.

The theoretical integration of the theories used in the past literature resulted in an integrated framework that will ease the identification and analysis of future research avenues (Alvesson & Sandberg, 2020). Future research can explore the interplay of significant theoretical constructs mentioned in the theoretical framework. Relationships defining AI-driven branding can be reimagined and restructured as advanced AI technologies and brand experiences collaborate. Further exploration of the inter-relationships emerging from the integrated framework can lead to an assessment of the best sustainable brand practices that brands can adopt in the age of futuristic technologies to persist and endure. The theoretical framework propels updated and emergent research in AI-driven branding for the futuristic advancement of the research domain.

Declaration of Competing Interest

We declare that there are no potential sources of conflict of interest. We confirm that no interest or relationship, financial or otherwise, might be perceived as influencing the author's objectivity.

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